

PH 504: Organic Optoelectronics

Course Code	PH 504
Course Name	Organic Optoelectronics
L-T-P-C	3-0-0-3
Approval	2nd Adhoc Senate (May 2012), in the Minor in Applied Physics curriculum

Course Contents

PART I

- **Organic Molecules**
- Electronic structure of atoms, Atomic and Molecular Orbitals, LCAO, Bonding and antibonding orbitals, Covalent Bond, Sigma and Pi Bonds, Energy Levels, Spectroscopic properties. [4 Lectures]
- **Photophysics of Molecules and Aggregates**
- Excited states: Absorption and emission, Singlet and triplet states, Radiative and non-radiative transitions, Aggregates, Vander Waals Bonding, Hydrogen Bonding, Dimer, Eximers. [2 Lectures]
- **Excitons**
- Wannier Exciton, Charge-transfer Exciton Frenkel Exciton, Exciton Diffusion, Excitonic Energy Transfer. [2 Lectures]
- **Conduction in Organic Solids**
- Conductivity: carrier concentration versus mobility, Carrier generation, Hopping transport, Mobility measurements, Traps. [2 Lectures]
- **Photovoltaics and Photodetectors**
- Photovoltaic Devices: Organic Heterojunction Photovoltaic Cells, Organic/Nanorod hybrid Photovoltaics, Gratzel Cells (Dye sensitized solar cells), Photodetector Devices. [5 Lectures]
- **Organic Light Emitting Devices**
- Basic OLED Properties, Charged Carrier Transport, Organic LEDs, Quantum Dot LEDs. [8 Lectures]
- **Lasing Action in Organic Semiconductors**
- Lasing Process, Optically Pumped Organic Lasers, Electrical Pumping of Organic Lasers. [2 Lectures]
- **Organic Thin Film Transistors**
- OFETs: Materials, Contacts, Applications, Nanotube Transistors. [2 Lectures]
- **Device Fabrication Technology**
- Growth Techniques: Evaporation, Langmuir-Biodget, Chemical Vapor Phase Deposition, Ink-Jet Printing, Self Assembly. [3 Lectures]

PART II

Project

- Literature review on a certain relevant topic. [10 Lectures]

Text Books

No textbook required. Lecture notes and handouts will be provided.

General References

1. Gilbert & Baggott, **Essentials of Molecular Photochemistry**, CRC Press, 1991.
2. K. K. Rohatgi, Mukherjee, **Fundamentals of Photochemistry**, New Age International, 1978.
3. Pope & Swenberg, **Electronic Processes of Organic Crystals and Polymers**, 2nd Edition, Oxford University press, 1999.
4. H. Meier, **Organic Semiconductors**, Verlag Chemie GmbH, 1974.
5. Wolfgang Briltting, **Physics of Organic Semiconductors**, John Wiley, 2005.